

Site-to-Site IPsec VPN Troubleshooting Using Pre-Shared Key

1. Objective

To configure and troubleshoot a site-to-site IPsec VPN between two routers in a simulated WAN. The pre-shared key was intentionally misconfigured on one router. The connectivity problem was then identified using the commands `show crypto isakmp sa` and `show run`, and the configuration was corrected.

2. Topology

The WAN simulation uses two routers connected through a serial WAN link. The internal networks are 192.168.1.0/24 on Router1 and 192.168.2.0/24 on Router0.

PC0/PC1/PC2 — Switch — Router1 ——— Serial WAN ——— Router0 — LAN/PC
192.168.1.0/24 200.1.1.2 200.1.1.1 192.168.2.0/24

3. IP Addressing

Device	Interface	IP Address	Purpose
Router1	G0/0/0	192.168.1.1/24	Inside/LAN
Router1	S0/1/0	200.1.1.2/30	WAN / VPN peer
Router0	S0/1/0	200.1.1.1/30	WAN / VPN peer
Router0	G0/0/0	192.168.2.1/24	Inside/LAN
PC0	Fa0	192.168.1.10/24	LAN host
PC1	Fa0	192.168.1.11/24	LAN host
PC2	Fa0	192.168.1.12/24	LAN host
Remote PC	Fa0	192.168.2.10/24	Remote LAN host

4. VPN Configuration

The following configuration was used for ISAKMP and IPsec.

Router1 Configuration

```
enable
configure terminal
crypto isakmp policy 10
  encryption aes
  authentication pre-share
  group 2
exit
crypto isakmp key cisco123 address 200.1.1.1
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.2.0 0.0.0.255
crypto map VPN-MAP 10 ipsec-isakmp
  set peer 200.1.1.1
  set transform-set VPN-SET
  match address 110
exit
interface serial 0/1/0
```

```
crypto map VPN-MAP
exit
```

Router0 Configuration

```
enable
configure terminal
crypto isakmp policy 10
  encryption aes
  authentication pre-share
  group 2
exit
crypto isakmp key cisco123 address 200.1.1.2
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
access-list 110 permit ip 192.168.2.0 0.0.0.255 192.168.1.0 0.0.0.255
crypto map VPN-MAP 10 ipsec-isakmp
  set peer 200.1.1.2
  set transform-set VPN-SET
  match address 110
exit
interface serial 0/1/0
  crypto map VPN-MAP
exit
```

5. Intentional Misconfiguration

To simulate a VPN connectivity fault, the pre-shared key on Router0 was intentionally changed from cisco123 to WRONG123.

```
Router0(config)# crypto isakmp key WRONG123 address 200.1.1.2
```

The resulting mismatch was:

- Router1 pre-shared key: cisco123
- Router0 pre-shared key: WRONG123
- Expected result: ISAKMP negotiation fails and the IPsec tunnel cannot be established.

6. Troubleshooting

6.1 Test Connectivity

Traffic was generated from the internal LAN toward the remote LAN:

```
PC0> ping 192.168.2.10
```

The observed ping showed one initial timeout followed by successful replies. This confirms basic IP reachability in the simulation, but ping success alone does not prove that the traffic is protected by IPsec.

6.2 Check ISAKMP Security Association

```
Router# show crypto isakmp sa
```

The captured output showed an empty IPv4 Crypto ISAKMP SA table. Therefore, no active ISAKMP security association had been established at that time.

6.3 Inspect the Running Configuration

```
Router# show run | section crypto
```

The running configuration was inspected to compare the ISAKMP pre-shared key and VPN parameters on both routers.

```
Router1:
crypto isakmp key cisco123 address 200.1.1.1
```

```
Router0 (after intentional fault):
crypto isakmp key WRONG123 address 200.1.1.2
```

The mismatch in the pre-shared key was identified as the configuration fault.

7. Corrective Action

The incorrect key on Router0 was replaced with the same key used by Router1:

```
Router0# configure terminal
Router0(config)# crypto isakmp key cisco123 address 200.1.1.2
Router0(config)# end
```

To remove any stale ISAKMP negotiation state, the existing ISAKMP security associations can be cleared:

```
Router0# clear crypto isakmp sa
```

8. Verification

Generate interesting traffic again:

```
PC0> ping 192.168.2.10
```

Then verify the VPN:

```
Router1# show crypto isakmp sa
Router1# show crypto ipsec sa
```

- A successful ISAKMP negotiation should show an established state such as QM_IDLE.
- show crypto ipsec sa should show IPsec security associations and increasing encapsulation/decapsulation counters.
- The remote LAN should remain reachable after the tunnel is established.

9. Result

The VPN troubleshooting exercise demonstrated how a pre-shared-key mismatch can prevent ISAKMP/IPsec tunnel establishment. The issue was located by comparing the output of show crypto isakmp sa and show run. The incorrect pre-shared key was corrected so that both VPN peers use the same key.

10. Commands Used / Quick Reference

```
show crypto isakmp sa
show run | section crypto
show crypto ipsec sa
clear crypto isakmp sa
```

This practical successfully covered IPsec VPN configuration, intentional pre-shared-key misconfiguration, diagnosis, and correction. The key troubleshooting lesson is that both VPN peers must use matching pre-shared keys and compatible ISAKMP/IPsec parameters. The commands `show crypto isakmp sa` and `show run` are useful for identifying tunnel negotiation and configuration problems.